INTERPRETABILITY AND THE BLACK HOLE INTERIOR

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Abstract. Neural networks simulate larger, interpretable networks using superposition. Black holes simulate the larger Hilbert space of their interior using superposition. Part I makes the resemblance precise: both are random compressions that act as approximate isometries on "thin" sets of states (low complexity for the black hole, sparsity for the network), and one theorem covers both. From it we build a dictionary between the physics of the interior (null states, state-dependent reconstruction, the Page transition, the Python's lunch) and feature recovery as compressed sensing, with an explicit list of where the dictionary is exact and where it is only analogy. We then test the randomness assumption on trained models: toy models of superposition, RoBERTa token embeddings and GloVe. Trained codes agree with Haar-random codes at second order—but a frame identity forces this for any tight frame—and differ in informative ways at fourth order, in anisotropy and in their tails, while still recovering sparse combinations at the Donoho–Tanner threshold. Part II asks what training does to the *orientation* of a code relative to a public frame, the property that monitoring depends on. In a toy sender–receiver model, a code that is legible in the public frame is a critical point of the training loss when its coordinates are unshared. Theory makes it a saddle whose curvature is proportional to the receiver's residual error; at our operating point that curvature is below 10^{-9}, and what moves the code is that adaptive optimizers turn vanishing gradients into finite steps. Accordingly, plain SGD on the sender alone preserves legibility at no cost to the task, while Adam spreads the code to no benefit. In superposition the legible code is not a critical point, drift follows the receiver, and the optimizer prescription fails. A quantitative test of the curvature theory was confounded by the receiver memorizing its training noise. Several claims from earlier drafts did not survive the experiments; the corrections are listed.

How these notes were made

The idea of comparing the non-isometric code of Akers, Engelhardt, Harlow, Penington and Vardhan [1] with superposition in neural networks, the decision to pursue its consequences for monitoring agents that communicate in activations, and the editorial direction of these notes are David Wakeham's. The derivations, code, experiments and most of the prose were produced by Claude (Anthropic; Claude Opus models) in conversation with D.W. between September and November 2026, with D.W. choosing directions, asking for checks, and editing. Every number in these notes is produced by a script in the accompanying repository (`bhi`; see Appendix A). Claims carry a status tag: [proved] (proof given here), [cited] (standard result, reference given), [checked numerically] (verified by the code but not proved), [heuristic] (derivation with uncontrolled steps), or [analogy]. Earlier versions of several Part II claims were wrong; Appendix B lists what changed and which experiment changed it. *Author's verification status:* at the time of writing D.W. has not yet independently checked every derivation; this sentence will be updated as he does.

Date: Version 0.1, November 2026. Written with Claude (Anthropic); see "How these notes were made".

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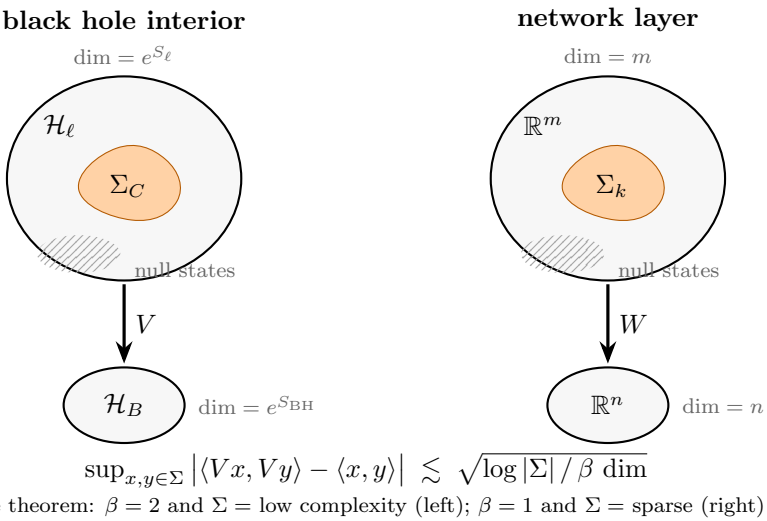


FIGURE 1. The two compressions. A large effective space is mapped by a linear code into a much smaller fundamental space. Most of the large space is annihilated (null states), yet on a thin set Σ—low complexity on the left, sparsity on the right—inner products survive. Theorem 3.6 is the statement that these are one phenomenon.

1. INTRODUCTION

Black holes, like neural networks, are hard to look inside: a black hole has a horizon; a neural network is polysemantic. We show this is more than a simile. In both cases, a large "effective" space (the black hole interior; feature space) is compressed by a linear map into a smaller "fundamental" space (microscopic degrees of freedom; physical neurons). This forces initially orthogonal directions to overlap, leading to null states and state-dependent operators in the black hole setting, and interference and polysemanticity in networks. We isolate a single condition for the validity of both effective descriptions: states or features form an appropriately "thin" set for which random linear compressions preserve inner products. We use this to build a correspondence between black holes and neural networks, make quantitative predictions about feature recovery, and test them in toy models of superposition and in the embeddings of trained language models. A collapsed star is a surprisingly rich guide to the inside of a neural network.

Of the parallels, the Python's lunch [8] deserves mention, because it concerns an *observer* rather than the code. A lunch is a region whose information is present in the fundamental description but whose reconstruction is exponentially complex. Its classical counterpart is the gap between statistical and computational recovery in sparse decoding. A monitor is also an observer—a decoder with a fixed frame and a bounded budget—and this is where Part II begins. For a black hole the code is a fixed fact about a state. For a network it is trained, so the question becomes dynamical: does optimization keep representations legible to a monitor, or move them away? Part II studies this in the smallest model where the question makes sense.

Summary of findings.

(1) **One theorem** [proved]. A Haar-random code over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ from dimension M to N preserves all inner products on a set Σ to accuracy δ once $\log|\Sigma| \ll \beta N \delta^2$ (Theorem 3.6). Low circuit complexity (AEHPV) and sparsity (the restricted isometry property from compressed sensing) are two special cases.

(2) **Exact moments** [proved] [checked numerically]. Overlap variances are $\frac{(M-N)(M-|c|^2)}{N(M^2-1)}$ over $\mathbb{C}$ and $\frac{(m-n)(m+(m-2)c^2)}{n(m-1)(m+2)}$ over $\mathbb{R}$; the real/complex difference is exactly the third Brauer invariant (Lemma 3.1).

(3) **Trained codes are not Haar, but are "Haar-good"** [checked numerically]. Second moments agree for a reason unrelated to randomness (Proposition 3.3). Toy-model codes are *better* than Haar at the extremes when features are plentiful, and maximally non-Haar (antipodal) when they are not. Real embeddings are anisotropic with heavy-tailed overlaps. Yet ℓ_1 recovery of random sparse combinations sits at the Haar threshold, as the known universality of the Donoho–Tanner transition [12] would lead one to expect (Section 5).

(4) **Legibility is a critical point, where there is room** [heuristic] [checked numerically]. In a sender–receiver toy, a code that is sparse in the public frame is a critical point in "leak" directions toward empty coordinates. Theory makes it a saddle with curvature proportional to the receiver's residual error (Section 7); at our operating point that curvature is below 10^{-9}, so the point is a plateau, and what matters is that Adam turns vanishing gradients into finite steps while gradient descent does not move. Accordingly, plain SGD on the sender alone keeps features on their coordinates at no cost to the task, even with Adam elsewhere—but only when coordinates are unshared. In superposition the legible code is not a critical point, drift follows the receiver, and the prescription fails. The quantitative curvature theory is untested in the full model (Section 8).

(5) **Corrections** [checked numerically]. Most of the frozen-monitor degradation reported in earlier drafts is *calibration* drift, repairable by refitting a scale and bias per feature. The residual directional drift is real but smaller. Raising task pressure did not spread the code, and putting the monitor in the loss did not open a hidden channel at this scale (Section 8, Appendix B).

Part I. The correspondence

2. Two compressions

2.1. **The black hole interior.** The literature on the quantum mechanics of black holes is enormous; the part we need concerns Hawking's information paradox and its recent resolution through holography [6, 7, 1]. In particular, we exploit the insight of Akers, Engelhardt, Harlow, Penington and Vardhan [1] that the effective description of the black hole interior—the "bulk" local physics, such as the creation and annihilation operators of quantum fields inside the horizon—is related to the fundamental description by a random linear compression. The fundamental description consists of the black hole's own remaining degrees of freedom, counted by its entropy, together with the radiation it has emitted; the compression acts on the interior modes alone and carries the radiation across untouched (Fig. 2). This single picture accounts for several pieces of black hole folklore: the Page curve of the radiation entropy, the state-dependence of interior operators, the absence of low-complexity null states, and the exponential cost of decoding the radiation.

Formally, we can forget about gravity and time evolution and focus on the linear algebra of inner-product spaces. The effective description of the interior is a Hilbert space $\mathcal{H}_\ell$ of dimension $|\ell| = e^{S_\ell}$; at late times S_ℓ exceeds the black hole's Bekenstein–Hawking entropy S_{BH}, because the interior keeps growing while the black hole shrinks. The fundamental description of the black hole is $\mathcal{H}_B$ with $|B| = e^{S_{\mathrm{BH}}}$. The two are related by

$$V\colon \mathcal{H}_\ell \to \mathcal{H}_B, \qquad V = \sqrt{|P|}\,\langle 0|_P\, U\, |\psi_0\rangle_f, \qquad |\ell||f| = |B||P|, \tag{1}$$

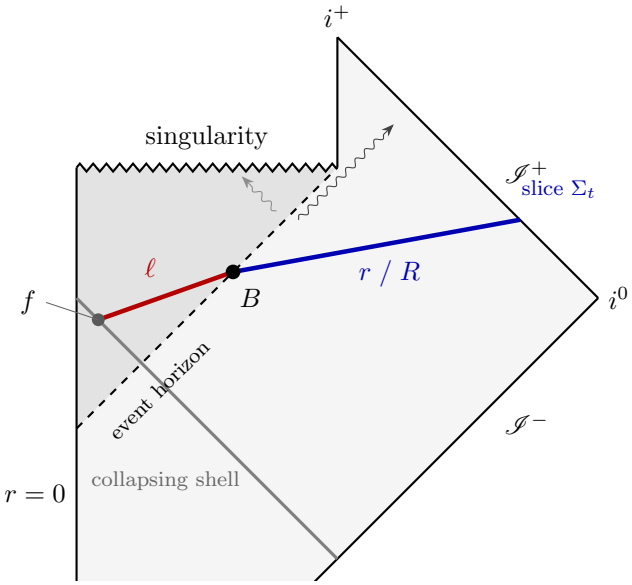


FIGURE 2. Late-time Cauchy slice Σ_t of an evaporating black hole formed by a collapsing shell (zoomed on the relevant region; null lines at 45°). Outside the horizon the slice carries the outgoing Hawking modes r, which are the same system R in both descriptions. Inside, it carries the infalling partner modes ℓ near the horizon and, at its inner end, the collapsed matter f, frozen in the state it fell in with. In the fundamental description the whole interior segment is encoded in the black hole's microstates B at the horizon crossing, with $\dim \mathcal{H}_B = e^{A/4G}$: the map V squeezes a segment onto a point whose size is its area. Lower-case labels are what a semiclassical observer assigns to a region; upper-case labels are the fundamental Hilbert spaces. They differ only at the horizon, and the difference is V.

extended by the identity on the radiation. Here U is a Haar-random unitary on $\mathcal{H}_\ell \otimes \mathcal{H}_f$ standing in for the unknown, scrambling dynamics; f is the collapsed matter, prepared in a fixed state $|\psi_0\rangle$ by the history of the collapse; and P is a bookkeeping remainder. The interior $\ell \otimes f$ has more states than the black hole B can hold, so the unitary must output them somewhere: P is that somewhere, discarded by projection onto $|0\rangle_P$—the states the black hole would have needed had it not shrunk.

The factor $\sqrt{|P|}$ makes typical states come out with unit norm. Indeed $\langle 0|_P U|\psi_0\rangle_f$ is a block of U, and $U^\dagger(I_B \otimes |0\rangle\langle 0|_P)U$ is a Haar-random projector of rank $|B|$ on a space of dimension $|\ell||f| = |B||P|$, so by Lemma 3.1, $\mathbb{E}\, V^\dagger V = |P| \cdot \frac{|B|}{|\ell||f|}\, I_\ell = I_\ell$.

For the linear algebra the ancillae are inessential and we use the *reduced model* $|f| = 1$:

$$V = \sqrt{M/N}\; P_N U, \qquad M = |\ell|,\ N = |B|, \tag{2}$$

$\sqrt{M/N}$ times N rows of a Haar unitary. Then $V^\dagger V = \frac{M}{N} Q$ with Q a Haar-random rank-N projector. Since $M > N$, $\dim\ker V \geq M - N$: almost all of the effective space consists of *null states* that the fundamental description cannot distinguish from zero. The content of [1] is that V is nonetheless an isometry to any observer of bounded complexity; Section 3 proves the version we need.

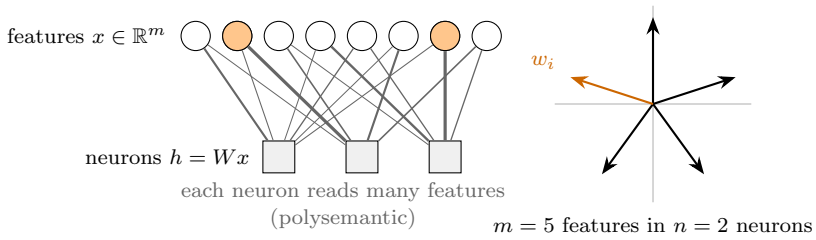


FIGURE 3. Superposition. Left: m features feed $n < m$ neurons; a sparse input (shaded) activates few features but every neuron reads several. Right: the classic toy-model solution with $m = 5$ features in $n = 2$ dimensions, a regular pentagon [2].

2.2. **Superposition.** The working hypothesis of mechanistic interpretability is that a trained network computes with *features*: directions in activation space that correspond to human-legible properties of the input. The obstacle is that features are not neurons. A neuron typically responds to several unrelated features and a feature is spread across many neurons—*polysemanticity*. Elhage et al. [2] showed that polysemanticity is what compression looks like from the inside. A network that must represent m features in $n < m$ neurons can, if the features are sparse, assign each a direction in $\mathbb{R}^n$ and accept that the directions overlap, because overlaps only matter when overlapping features are simultaneously active. This is *superposition*: the network simulates a larger, monosemantic network (Fig. 3).

The effective description is feature space $\mathbb{R}^m$; a feature activation x is nonzero on few coordinates, and we write Σ_k for the unit vectors with at most k nonzero entries. The fundamental description is the activation space $\mathbb{R}^n$. The two are related by $W\colon \mathbb{R}^m \to \mathbb{R}^n$, $h = Wx$, whose columns $w_1, \dots, w_m$ are the feature directions. In the toy models of [2] the readout is $\hat{x} = \mathrm{ReLU}(W^\top h + b)$. The transpose alone cannot undo superposition (any linear map $\mathbb{R}^n \to \mathbb{R}^m$ has rank $\le n$); $W^\top W x = x + e$ is correct on the active coordinates up to the *interference* $e_i = \sum_{j \neq i} \langle w_i, w_j \rangle x_j$, and the ReLU with negative bias projects back onto the sparse set by thresholding the interference away (Proposition 4.2). Since $m > n$, W cannot be an isometry; its kernel contains the dense activation patterns that a sparse network never produces.

2.3. **The static dictionary.** Table 1 lines up the two settings. The quantitative content of "preserves inner products on a distinguished set" is the subject of Section 3. The one structural asymmetry is that V is assembled from a unitary with ancillae, because in gravity the scrambling is unitary and the compression has to be manufactured from it, whereas W is trained directly; nothing in the dictionary depends on the ancillae.

3. THE UNIFYING THEOREM

Throughout, a code maps an M-dimensional effective space to an N-dimensional fundamental space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$; write $\beta = \dim_{\mathbb{R}} \mathbb{F}$. The *Haar code* is $V = \sqrt{M/N}\, P_N U$ with U Haar on $U(M)$ (complex) or $O(M)$ (real); in the real case we write W, m, n. For a set Σ of unit vectors the distortion is $\epsilon_\Sigma(V) = \sup_{x,y \in \Sigma} |\langle Vx, Vy \rangle - \langle x, y \rangle|$.

3.1. **Second moments.**

	black hole (AEHPV)	network (superposition)
effective space	interior $\mathcal{H}_\ell$, $\dim = e^{S_\ell}$	features $\mathbb{R}^m$
fundamental space	microstates $\mathcal{H}_B$, $\dim = e^{S_{\mathrm{BH}}}$	neurons $\mathbb{R}^n$
code	V, from a Haar unitary	W, trained (Haar as baseline)
null space	$\dim \geq M - N$, all high complexity	$\ker W$, no k-sparse vectors
distinguished set	complexity $\leq C$	k-sparse
its size	$\log\lvert\Sigma_C\rvert \lesssim C \log S_\ell$	$\log \binom{m}{k} \leq k \log(em/k)$
overlap fluctuations	$\approx e^{-S_{\mathrm{BH}}/2}$	$\approx n^{-1/2}$
field	$\mathbb{C}$ ($\beta = 2$)	$\mathbb{R}$ ($\beta = 1$)

TABLE 1. The static dictionary. The exponential in the left column becomes polynomial on the right: $e^{S_{\mathrm{BH}}} \leftrightarrow n$, hence $S_{\mathrm{BH}} \leftrightarrow \log n$.

Lemma 3.1 (Projector moments)**.** *[proved]Let Q be a Haar-random rank-N orthogonal projector on $\mathbb{F}^M$. Then $\mathbb{E}Q = \frac{N}{M} I$, and*

$$\mathbb{F} = \mathbb{C}: \quad \mathbb{E}[Q \otimes Q] = aI + bF, \qquad a = \tfrac{N(NM-1)}{M(M^2-1)}, \quad b = \tfrac{N(M-N)}{M(M^2-1)};$$

$$\mathbb{F} = \mathbb{R}: \quad \mathbb{E}[Q \otimes Q] = \alpha I + \gamma(F + \Omega), \qquad \gamma = \tfrac{N(M-N)}{M(M-1)(M+2)}, \quad \alpha = \tfrac{N}{M} - \gamma(M+1),$$

where F is the swap and $\Omega = \lvert\omega\rangle\langle\omega\rvert$ with $\lvert\omega\rangle = \sum_i e_i \otimes e_i$.

Proof. $\mathbb{E}Q$ commutes with every unitary (orthogonal) g, since $g^\dagger Q g$ has the same law as Q; so $\mathbb{E}Q = cI$ and tr gives $c = N/M$. Likewise $\mathbb{E}[Q \otimes Q]$ commutes with all $g \otimes g$. Over $\mathbb{C}$ the commutant of $\{g \otimes g\}$ is spanned by I, F (Schur–Weyl); over $\mathbb{R}$ by I, F, Ω (Brauer). Fix the coefficients by traces. Over $\mathbb{C}$: $\operatorname{tr}\mathbb{E}[Q \otimes Q] = (\operatorname{tr} Q)^2 = N^2 = aM^2 + bM$ and $\operatorname{tr}(F\,\mathbb{E}[Q \otimes Q]) = \operatorname{tr} Q^2 = N = aM + bM^2$, which solve to the stated a, b. Over $\mathbb{R}$ there is a third equation, $\operatorname{tr}(\Omega\,\mathbb{E}[Q \otimes Q]) = \operatorname{tr}(QQ^\top) = N$; using $\operatorname{tr} F = \operatorname{tr}\Omega = M$, $\operatorname{tr}(F\Omega) = M$, $\operatorname{tr} F^2 = \operatorname{tr}\Omega^2 = M^2$, the second and third equations are symmetric in the coefficients of F and Ω, so these are equal ($= \gamma$), and $N^2 = \alpha M^2 + 2\gamma M$, $N = \alpha M + \gamma(M^2 + M)$ give $\gamma M(M-1)(M+2) = N(M-N)$. □

Proposition 3.2 (Exact overlap variances)**.** *[proved] [checked numerically]For fixed unit vectors with overlap c, $\mathbb{E}\langle Vx, Vy\rangle = c$ and*

$$\mathrm{Var}_{\mathbb{C}} = \frac{(M-N)(M-\lvert c\rvert^2)}{N(M^2-1)}, \qquad \mathrm{Var}_{\mathbb{R}} = \frac{(m-n)\left(m + (m-2)c^2\right)}{n\,(m-1)(m+2)}, \qquad \mathrm{Var}_{\mathrm{Gauss}} = \frac{1+c^2}{n},$$

the last for i.i.d. $\mathcal{N}(0, 1/n)$ entries. For $N \ll M$ the complex variance is $\approx 1/N$ independently of c; the real one is $\approx (1 + c^2)/n$, and the extra c^2 is exactly the Ω term of Lemma 3.1, the Brauer invariant that is present for $O(n)$ and absent for $U(n)$. Haar frames beat Gaussian ones by a factor $\approx (1 - n/m)$.

Proof. $\mathbb{E}\lvert\langle x\rvert Q \lvert y\rangle\rvert^2 = (\langle x\rvert \otimes \langle y\rvert)\mathbb{E}[Q \otimes Q](\lvert y\rangle \otimes \lvert x\rangle) = a\lvert c\rvert^2 + b$ over $\mathbb{C}$; over $\mathbb{R}$, $\mathbb{E}(x^\top Q y)^2 = (\alpha + \gamma)c^2 + \gamma$. Multiply by $(M/N)^2$, subtract $\lvert c\rvert^2$, simplify. For Gaussian entries apply Isserlis to the n independent terms $\langle g_i, x\rangle\langle g_i, y\rangle$. □

Proposition 3.3 (Frame identity)**.** *[proved]Let $w_1, \ldots, w_m$ be unit vectors in $\mathbb{R}^n$ forming a tight frame ($WW^\top = \frac{m}{n} I$). Then*

$$\frac{1}{m(m-1)} \sum_{i \neq j} \langle w_i, w_j\rangle^2 = \frac{m-n}{n(m-1)}.$$

Hence every *unit-norm tight frame agrees with the Haar prediction $\frac{m(m-n)}{n(m-1)(m+2)}$ for the mean-square interference, up to a factor $m/(m+2)$.*

Proof. $\sum_{i,j}\langle w_i, w_j\rangle^2 = \|W^\top W\|_F^2 = \|WW^\top\|_F^2 = (m/n)^2 n$; subtract the m diagonal terms. □

This matters for experiments: agreement with Haar at second order is evidence of tightness, not of randomness.

3.2. **Concentration and the theorem.**

Lemma 3.4 (Law of a compressed norm)**.** *[proved]For fixed unit z, $\|Vz\|^2 \overset{d}{=} \frac{M}{N}\cdot\frac{A}{A+B}$ with $A\sim\chi^2_{\beta N}$, $B\sim\chi^2_{\beta(M-N)}$ independent.*

Proof. $\|Vz\|^2 = \frac{M}{N}\|P_N U z\|^2$ and Uz is uniform on the unit sphere of $\mathbb{F}^M\cong\mathbb{R}^{\beta M}$, i.e. $g/\|g\|$ with g standard Gaussian. $\|P_N g\|^2\sim\chi^2_{\beta N}$ and $\|(1-P_N)g\|^2\sim\chi^2_{\beta(M-N)}$ independently. □

Lemma 3.5 (χ^2 concentration)**.** *[proved]For $Z\sim\chi^2_k/k$ and $0<s\le 1$: $\mathbb{P}[|Z-1|\ge s]\le 2e^{-ks^2/12}$.*

Proof. Chernoff with $\mathbb{E}e^{\lambda\xi^2} = (1-2\lambda)^{-1/2}$. Upper tail: $\lambda=\frac{s}{2(1+s)}$ gives exponent $-\frac{k}{2}[s-\log(1+s)]\le -\frac{k}{2}(\frac{s^2}{2}-\frac{s^3}{3})\le -\frac{ks^2}{12}$. Lower tail: $\lambda=\frac{s}{2(1-s)}$ for the variable $-\xi^2$ gives exponent $-\frac{k}{2}[-s-\log(1-s)]\le -\frac{ks^2}{4}$. □

Theorem 3.6 (Random codes are isometries on thin sets)**.** *[proved]Let V be the Haar code over $\mathbb{F}$ and T a finite set of unit vectors. For $0<\delta\le 1$,*

$$\mathbb{P}\Big[\sup_{z\in T}\big|\|Vz\|^2-1\big|>\delta\Big]\le 4|T|\,e^{-\beta N\delta^2/108}.$$

Consequently, for any finite set Σ of unit vectors, $\epsilon_\Sigma(V)\le 2\delta$ with probability at least $1-16|\Sigma|^2 e^{-\beta N\delta^2/108}$; that is, the code is an approximate isometry on Σ as soon as

$$\boxed{\log|\Sigma| \;\ll\; \beta\, N\,\delta^2} \tag{3}$$

Proof. By Lemma 3.4, $\|Vz\|^2 = \frac{A/\beta N}{(A+B)/\beta M}$. If numerator and denominator are both within $1\pm\delta/3$ the ratio is within $1\pm\delta$ (since $\frac{1+t}{1-t}\le 1+3t$ for $t\le\frac13$), and by Lemma 3.5 each fails with probability at most $2e^{-\beta N\delta^2/108}$. Union-bound over T. For the second statement apply this to $T=\{(x+\omega y)/\|x+\omega y\|\}$ with $\omega\in\{\pm1\}$ (real) or $\{\pm1,\pm i\}$ (complex), $|T|\le 4|\Sigma|^2$, and use polarization: $|\langle Vx,Vy\rangle-\langle x,y\rangle|\le\frac14\sum_\omega\delta\|x+\omega y\|^2\le 2\delta$. □

The theorem never asks which field or which set it is given; it sees only β and $\log|\Sigma|$. Its two specializations are the two halves of the correspondence.

Corollary 3.7 (AEHPV: low complexity)**.** *[proved]Over $\mathbb{C}$ with $N=e^{S_{\rm BH}}$, let Σ_C be the states preparable from a fixed reference state by circuits of at most C gates from a finite gate set G acting on the S_ℓ interior qubits. Then $|\Sigma_C|\le(|G|S_\ell^2)^C$, and with high probability*

$$\sup_{\psi,\phi\in\Sigma_C}\big|\langle\psi|V^\dagger V|\phi\rangle-\langle\psi|\phi\rangle\big|\lesssim\sqrt{C\log S_\ell}\;e^{-S_{\rm BH}/2},$$

negligible for all C sub-exponential in $S_{\rm BH}$. In particular no state of sub-exponential complexity is null.

Corollary 3.8 (Restricted isometry)**.** *[proved]Over $\mathbb{R}$, W satisfies $(1-\delta)\le\|Wx\|^2\le(1+\delta)$ for all k-sparse unit x with high probability once $n\gtrsim\delta^{-2}k\log(em/k)$.*

Proof. For each support S take an η-net of the unit sphere of $\mathbb{R}^S$ with at most $(3/\eta)^k$ points; the union over $\binom{m}{k}$ supports has $\log|T|\le k\log(em/k)+k\log(3/\eta)$. Apply Theorem 3.6 to the net, then extend from the net to the sphere: if $1-\delta'\le\|Wz\|\le 1+\delta'$ on the net and $A=\sup\|Wz\|$, then $A\le 1+\delta'+\eta A$, so $A\le(1+\delta')/(1-\eta)$, and similarly from below; take $\eta=\delta'/4$. □

Remark 3.9 (No simple null vectors). If $\delta_{2k} < 1$ then no nonzero $2k$-sparse vector is in $\ker W$, so $\operatorname{spark}(W) > 2k$ and each k-sparse vector is determined by its image. The dictionary entry is: $\operatorname{spark}(W)$ is the complexity of the cheapest null state.

Remark 3.10 (Gaussian width). [cited]The finite-set condition $\log|\Sigma| \ll \beta N \delta^2$ is the special case of Gordon's theorem [13, 14]: a Gaussian $n \times m$ map is an approximate isometry on any $T \subset S^{m-1}$ with distortion $\lesssim w(T)/\sqrt{n}$, where $w(T) = \mathbb{E} \sup_{z \in T} \langle g, z \rangle$. For finite sets $w(T) \le \sqrt{2 \log |T|}$ (Jensen on $\mathbb{E} \max e^{\lambda \langle g, z \rangle}$), and for the k-sparse sphere $w(\Sigma_k) \le \sqrt{k} + \sqrt{2 \log \binom{m}{k}}$ (the same argument applied to the 1-subgaussian variables $\|g_S\| - \mathbb{E}\|g_S\|$, with $\mathbb{E}\|g_S\| \le \sqrt{k}$). Width is the right size for Σ; low complexity and sparsity are two ways of having small width. We do not need the complex version of Gordon's theorem, because both corollaries above go through the finite-set form.

Remark 3.11 (Where the exponential goes). Gravity compares $\operatorname{poly}(C)$ with $e^{S_{\mathrm{BH}}}$; a network compares $k \log(m/k)$ with n. The dictionary is $e^{S_{\mathrm{BH}}} \leftrightarrow n$, so corrections that are $e^{-S/2}$ in gravity are $n^{-1/2}$ in a network, and an effective description that is quasi-exact for a black hole is merely approximate for a transformer. This is the largest quantitative deflation in passing from one column of Table 1 to the other.

4. The reconstruction dictionary

4.1. **Decoders.**

On both sides the naive decoder is the adjoint of the encoder: the Petz (transpose) map [9] in gravity, $W^\top$ followed by a nonlinearity in a network.

Lemma 4.1. *[proved]If $\epsilon_\Sigma(V) \le \epsilon$ then $|\langle y, V^\dagger V x \rangle - \langle y, x \rangle| \le \epsilon$ for all $x, y \in \Sigma$.*

That is all one can say: $V^\dagger V x$ agrees with x against every test vector in Σ and may be badly wrong elsewhere. In a network this "elsewhere" is the interference, and the nonlinearity's job is to project back to Σ.

Proposition 4.2 (One-step thresholding)**.** *[proved]If the columns of W are unit vectors with coherence $\mu = \max_{i \ne j} |\langle w_i, w_j \rangle|$ and $x \in \{0,1\}^m$ has at most k ones, then $|(W^\top W x)_i - x_i| \le \mu k$; so $\mathrm{ReLU}(W^\top W x - \frac{1}{2})$ followed by rounding recovers x when $\mu k < \frac{1}{2}$.*

For random unit columns $\mu \approx \sqrt{2 \log m / n}$, so one step tolerates $k \lesssim \sqrt{n / \log m}$, whereas the isometry holds up to $k \sim n / \log(m/k)$; iterative and convex decoders close the gap (Section 4.3).

4.2. **State dependence.**

Proposition 4.3 (Why reconstruction must be state-dependent)**.** *[proved]In the reduced model, for any operator A on the effective space,*

$$\mathbb{E}\big[V^\dagger V \, A \, V^\dagger V\big] = \frac{M(NM-1)}{N(M^2-1)} \, A + \frac{M(M-N)}{N(M^2-1)} \, \mathrm{tr}(A) \, I \;\approx\; A + \frac{\mathrm{tr}\, A}{N} \, I.$$

So the state-independent reconstruction $VAV^\dagger$ has, in every state $V|\phi\rangle$, an expectation biased by $\approx e^{-S_{\mathrm{BH}}} \mathrm{tr}\, A$. For $A = I$ the reconstruction $VV^\dagger$ has expectation exactly $\frac{M}{N} \langle \phi | V^\dagger V | \phi \rangle \approx e^{S_\ell - S_{\mathrm{BH}}}$: the fundamental description of the identity is not even approximately the identity. A state-dependent reconstruction A_ϕ defined by $A_\phi V|\phi\rangle = VA|\phi\rangle$ inherits the accuracy of Corollary 3.7 whenever $|\phi\rangle$, $A|\phi\rangle$ and $B|\phi\rangle$ are low-complexity.

Proof. Index computation with Lemma 3.1: $\mathbb{E}[QAQ]_{ij} = \sum_{kl} \mathbb{E}[Q_{ik} Q_{lj}] A_{kl} = a A_{ij} + b \, \delta_{ij} \, \mathrm{tr}\, A$; multiply by $(M/N)^2$. For $A = I$ use $(V^\dagger V)^2 = \frac{M}{N} V^\dagger V$. The last statement is $\langle \phi | V^\dagger B_\phi^\dagger A_\phi V | \phi \rangle = \langle VB\phi, VA\phi \rangle$ and Corollary 3.7. □

The mechanism is instructive. $\Delta = V^\dagger V - I$ has small matrix elements between low-complexity states, but $\|\Delta\phi\|^2 \approx M/N - 1$ is huge: the error vector points entirely in high-complexity directions, which no low-complexity test vector sees. $VAV^\dagger$ sandwiches A between two such vectors. The network analogue is a decoder fitted on one distribution over Σ, which fails silently on another.

4.3. **The Page transition and the Donoho–Tanner transition.**

Proposition 4.4 (Lubkin; Rényi-2 Page curve)**.** *[proved] [checked numerically]For a Haar-random pure state on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$, $\mathbb{E}\,\mathrm{tr}\,\rho_A^2 = \frac{d_A+d_B}{d_A d_B+1}$.*

Proof. $|\Psi\rangle\langle\Psi|$ is a Haar rank-one projector, so $\mathbb{E}|\Psi\rangle\langle\Psi|^{\otimes 2} = \frac{I+F}{M(M+1)}$ by Lemma 3.1; $\mathrm{tr}\,\rho_A^2 = \mathrm{tr}[(F_A \otimes I)|\Psi\rangle\langle\Psi|^{\otimes 2}]$ and the two traces give $d_A d_B^2 + d_A^2 d_B$. □

Subsystem A is maximally mixed until it is more than half the system: the transition sits at $\log d_A = \log d_B$, an equality of two entropies, which in the gravitational description is the crossover of two quantum extremal surfaces [6, 7].

On the classical side, basis pursuit $\hat{x} = \arg\min\{\|z\|_1 : Wz = Wx\}$ returns x if and only if $\ker W \cap D(\|\cdot\|_1, x) = \{0\}$, where D is the descent cone [proved]. A random kernel misses a fixed cone with overwhelming probability when $n > \delta(D)$ and hits it when $n < \delta(D)$, with a window of width $O(\sqrt{m})$ [11] [cited]. The statistical dimension is computable:

Proposition 4.5 (Location of the transition)**.** *[proved] [checked numerically]For x with $k = \rho m$ nonzero entries, $\delta(D)/m \le \psi(\rho) = \inf_{\tau\ge 0}\{\rho(1+\tau^2) + (1-\rho)\,2[(1+\tau^2)\Phi(-\tau) - \tau\varphi(\tau)]\}$, tight up to $O(1/\sqrt{m})$. For nonnegative x recovered by $\min\{\sum z : Wz = Wx,\ z \ge 0\}$ the factor 2 is absent (only the upper tail of the off-support coordinates contributes), giving $\psi_+(\rho)$.*

Proof. $D^\circ = \mathrm{cone}(\partial f(x))$, so $\delta(D) = \mathbb{E}\,\mathrm{dist}(g, \mathrm{cone}\,\partial f)^2 \le \inf_\tau \mathbb{E}\,\mathrm{dist}(g, \tau\partial f)^2$. For $\|\cdot\|_1$ the distance splits coordinatewise into $(g_i - \tau\,\mathrm{sign}\,x_i)^2$ on the support and $(|g_i| - \tau)_+^2$ off it; $\mathbb{E}(g-\tau)_+^2 = (1+\tau^2)\Phi(-\tau) - \tau\varphi(\tau)$. For the nonnegative problem the subdifferential off the support is $\{u \le 1\}$, so only $(g_i - \tau)_+^2$ appears. □

Solving $m\psi(k/m) = n$ gives the Donoho–Tanner curve [10]. In our code, $n = 40$, $m = 120$ gives $k^* = 12.2$ (signed) and 17.2 (nonnegative); simulated 50% crossings are at $k \approx 12.5$ and 18.

Dictionary entry 4.6 (Page ↔ Donoho–Tanner; QES crossover ↔ ALMT edge)**.** [analogy] (structural) Both are typicality statements about a Haar or Gaussian object, both are sharp in large dimension, and both are located by an equality between two dimension-like functionals—one measuring what is available, one measuring what must be resolved. What the parallel is *not*: there is no extremization principle on the classical side. The QES prescription minimizes over candidate surfaces; $\delta(D)$ is a single number attached to a cone.

4.4. **The Python's lunch and the computational–statistical gap.** Every k-sparse x is determined by Wx iff $\mathrm{spark}(W) > 2k$, and a Gaussian W has spark $= n + 1$ almost surely, so $n \ge 2k$ suffices [proved]. The decoder this implies is ℓ_0 minimization, NP-hard in general [16]; the tractable ℓ_1 decoder needs $n \gtrsim 2k\log(em/k)$. In the window between, x is determined by Wx but no efficient decoder is known (for Gaussian W the hardness is conjectural).

Dictionary entry 4.7 (Python's lunch ↔ computational–statistical gap)**.** [analogy]A lunch is a region encoded in the boundary but whose reconstruction costs exponentially in the area difference between the minimal and bulge surfaces [8]. In both cases the obstruction is not information-theoretic, and in both the *existence* of the obstruction is visible from outside: the area difference is a boundary quantity, and the failure of ℓ_1 to produce a small residual is computable.

4.5. **Cloning.**

Dictionary entry 4.8 (Complementarity ↔ nothing)**.** [proved] (disanalogy) If the radiation can reconstruct an interior mode, the remaining black hole cannot (monogamy). Classically, $x \mapsto (x, x)$ is linear, and any number of parties can decode the same activation. The only obstructions to reading a classical code are ignorance of the code and computational cost—never a law of nature.

This is the disanalogy with the largest practical consequence, and it runs in a comforting direction for monitoring.

gravity	sparse coding	where	status
non-isometric V	compression W	§2	definition
$V^\dagger V \approx I$ on low complexity	restricted isometry on Σ_k	Theorem 3.6	proved
$e^{-S_{\mathrm{BH}}/2}$ corrections	$n^{-1/2}$ corrections	Proposition 3.2	proved, checked
no low-complexity null states	$\mathrm{spark}(W) > 2k$	Corollary 3.8	proved
Petz reconstruction	$W^\top$ then threshold	Proposition 4.2	proved
state-dependent operators	decoders valid only on Σ	Proposition 4.3	proved
Page transition	Donoho–Tanner transition	Proposition 4.5	proved, checked
QES crossover	ALMT edge $n = \delta(D)$	§4.3	structural analogy
Python's lunch	computational–statistical gap	§4.3	analogy; conjectural
complementarity	—		disanalogy
area, RT, evaporation dynamics	—		no counterpart

TABLE 2. The reconstruction dictionary with the standing of each entry.

5. Experiments: are trained codes Haar-random?

The theorem is about Haar codes; networks are trained. We test four things: the exact formulas (E01), trained toy models of superposition (E02), and the token and word embeddings of two trained language models (E03).

5.1. **E01: the formulas.** Monte Carlo over 3000 draws confirms Proposition 3.2 for $M = 64$ and $N \in \{4, 16, 48\}$ and the Lubkin formula (Table 3). The real/complex difference is visible in two places: the c^2 dependence of the real variance, and the fourth moment of the interference, which is 3 (real Gaussian) against 2 (complex Gaussian, for $|\cdot|$) once N is not tiny.

5.2. **E02: toy models of superposition.** We train the ReLU-output model of [2] with $n = 32$ neurons, uniform feature importance, and $m \in \{64, 128, 256\}$ features at several sparsities, then compare the represented feature directions (columns of norm > 0.5) with Haar frames of the same shape (Table 4, Fig. 4). Three findings. (i) Every trained code is an essentially exact tight frame, so by Proposition 3.3 its second moment *must* match Haar; it does, to three digits. (ii) Beyond second order the trained codes are not Haar. With $m = 2n$ they are n antipodal pairs—excess kurtosis ≈ 60, coherence 1—a highly structured code that is ideal for nonnegative features and a ReLU readout and terrible for signed ones. With $m = 8n$ they are *better* than Haar at the extremes: negative excess kurtosis and lower coherence. Training finds structured

			complex Haar		real Haar		Gaussian		4th moment	
M	N	c	MC	exact	MC	exact	MC	exact	real	complex
64	4	0.00	0.2474	0.2344	0.2460	0.2309	0.2701	0.2500	3.96	2.37
64	4	0.60	0.2343	0.2331	0.3284	0.3114	0.3438	0.3400	5.02	2.73
64	4	0.95	0.2485	0.2311	0.4308	0.4327	0.4839	0.4756	5.36	4.94
64	16	0.00	0.0472	0.0469	0.0477	0.0462	0.0641	0.0625	3.05	2.03
64	16	0.60	0.0470	0.0466	0.0620	0.0623	0.0790	0.0850	3.12	2.10
64	16	0.95	0.0468	0.0462	0.0882	0.0865	0.1160	0.1189	3.00	2.85
64	48	0.00	0.0054	0.0052	0.0050	0.0051	0.0200	0.0208	3.03	2.02
64	48	0.60	0.0053	0.0052	0.0071	0.0069	0.0280	0.0283	3.02	2.28
64	48	0.95	0.0054	0.0051	0.0094	0.0096	0.0382	0.0396	3.07	2.92

TABLE 3. E01: variance of $\langle Vx, Vy\rangle$ over 3000 draws against Proposition 3.2; last columns: normalized fourth moment of the deviation (3 for a real Gaussian, 2 for $|\cdot|$ of a complex one). Lubkin purity, Monte Carlo (exact), $d_B = 16$: $d_A = 4$: 0.3077 (0.3077); $d_A = 16$: 0.1250 (0.1245); $d_A = 64$: 0.0781 (0.0780).

			mean $\langle w_i, w_j\rangle^2$		excess kurt.		max positive		tight-		recovery at k		TMS loss	
m	p	rep.	trained	Haar	trained	Haar	trained	Haar	ness	k (k^*)	trained	Haar	trained	Haar
64	0.1	64	0.0159	0.0163	59.97	-0.00	0.006	0.414	1.000	18 (17.9)	0.03	0.67	0.106	0.491
64	0.03	64	0.0159	0.0163	59.96	0.02	0.010	0.485	1.000	18 (17.9)	0.07	0.47	0.010	0.090
128	0.03	128	0.0237	0.0240	1.31	-0.10	0.165	0.554	0.997	12 (11.7)	0.47	0.47	0.206	0.431
128	0.01	128	0.0238	0.0240	1.10	-0.12	0.149	0.548	0.996	12 (11.7)	0.40	0.50	0.027	0.095
256	0.01	256	0.0275	0.0276	-0.63	-0.15	0.239	0.636	0.999	8 (8.4)	0.50	0.77	0.188	0.340
256	0.003	256	0.0275	0.0277	-0.68	-0.14	0.236	0.603	0.998	8 (8.4)	0.60	0.77	0.025	0.072

TABLE 4. E02: toy models of superposition ($n = 32$) against Haar frames of the same shape. "rep." = represented features; tightness = participation ratio of the frame operator/n (1 = tight). Recovery: nonnegative ℓ_1, 30 trials at the k nearest the predicted threshold k^*. TMS loss: trained model vs a Haar frame with the best uniform bias. The mean overlap of every trained code equals $-1/(m-1)$ to four digits: its feature directions sum to zero.

codes, as engineers do; the random code is the floor. (iii) On the task they were trained for, trained codes beat the best uniformly-biased Haar frame by large factors, yet their ℓ_1 recovery thresholds are close to the Haar threshold and to the statistical-dimension prediction, consistent with the universality of that threshold [12].

5.3. **E03: language-model embeddings.** We extracted the RoBERTa-base [23] token embedding matrix (m = 50,261 tokens, $n = 768$) and GloVe 100d [24] (m = 400,000, $n = 100$) and treated vocabulary items as features. Each is analysed raw, centred (mean removed) and whitened (multiplied by $\Sigma^{-1/2}$, which forces a tight frame) (Table 5, Fig. 5). Real embeddings depart from Haar in two layers. The first is *anisotropy*: a common mean direction and a spectrum far from flat, so the frame is far from tight and the mean-square overlap is several times the Haar value (a known property of language-model embeddings [21]). Whitening removes this by construction and returns the second moment to $1/n$ (again by Proposition 3.3). The second

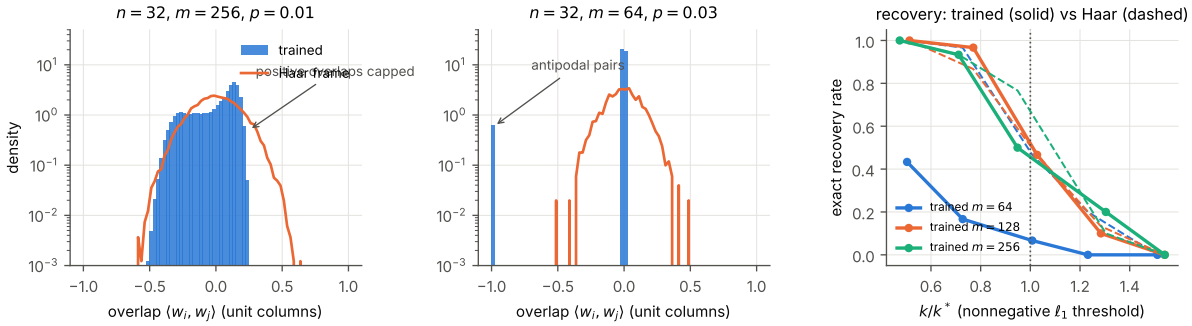


FIGURE 4. E02. Left: overlaps between trained feature directions (bars) and between Haar-frame directions (line) for $m = 256$, $p = 0.01$; trained codes are platykurtic, with fewer large overlaps. Middle: $m = 64$; the trained code is 32 antipodal pairs. Right: exact recovery of nonnegative k-sparse combinations by ℓ_1 minimization, trained vs Haar, with the predicted threshold k^* (dashed).

model	version	$n\,\mathbb{E}\cos^2$ (Haar ≈ 1)	excess kurt.	t	$\mathbb{P}(\lvert\cos\rvert > t)$ observed	$\mathbb{P}(\lvert\cos\rvert > t)$ Gaussian	max coherence observed	max coherence Haar	tightness
RoBERTa	raw	2.70	2.77	0.2	4.9e-03	3.0e-08	0.81	0.17	0.35
	centred	2.13	3.05	0.2	2.5e-03	3.0e-08	0.81	0.17	0.43
	whitened	0.99	5.09	0.2	2.9e-04	3.0e-08	0.88	0.17	0.85
GloVe	raw	4.77	-0.26	0.5	1.7e-02	5.7e-07	0.96	0.55	0.20
	centred	1.53	0.64	0.5	9.1e-04	5.7e-07	0.98	0.55	0.65
	whitened	1.02	0.42	0.5	2.0e-04	5.7e-07	0.94	0.55	0.96

model	m_{sub}	k (k^*)	exact ℓ_1 recovery rate centred	exact ℓ_1 recovery rate whitened	exact ℓ_1 recovery rate Haar
RoBERTa	1536	222 (296)	1.00	1.00	1.00
RoBERTa	1536	296 (296)	0.50	0.50	0.62
RoBERTa	1536	370 (296)	0.00	0.00	0.00
GloVe	200	19 (39)	1.00	1.00	1.00
GloVe	200	29 (39)	1.00	1.00	0.97
GloVe	200	35 (39)	0.83	0.93	0.83
GloVe	200	39 (39)	0.57	0.47	0.43
GloVe	200	42 (39)	0.23	0.27	0.30
GloVe	200	48 (39)	0.00	0.00	0.03

TABLE 5. E03: language-model embeddings as codes. Top: overlap statistics over 4×10^5 random pairs; t is the tabulated threshold nearest $4/\sqrt{n}$ ($5.5/\sqrt{n}$ for RoBERTa, $5/\sqrt{n}$ for GloVe); the Gaussian column is the tail of $\mathcal{N}(0, 1/n)$; coherence over 4000 random items. Bottom: signed ℓ_1 recovery from a random sub-dictionary of $m_{\text{sub}} = 2n$ items (k^*: predicted threshold; RoBERTa 8 trials per entry, GloVe 30).

layer survives whitening: the *tails* of the overlap distribution are far heavier than Gaussian. Listing the most coherent pairs (`scripts/e03_top_pairs.py`) shows what they are. In RoBERTa

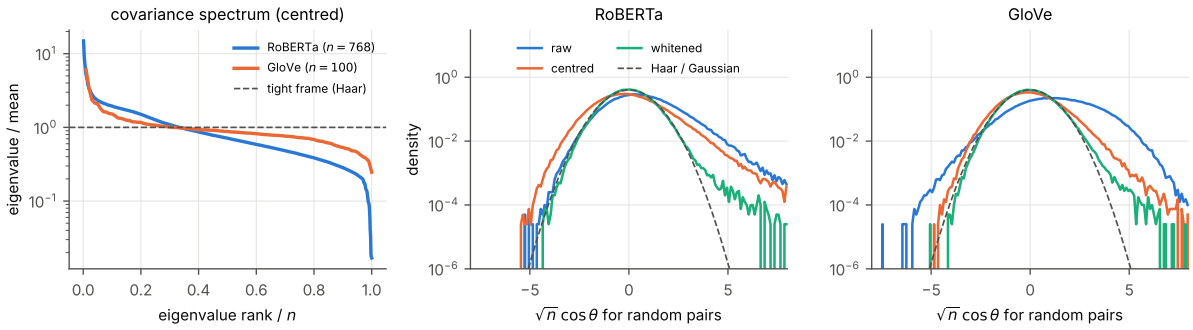


FIGURE 5. E03. Left: covariance spectra of centred embeddings, normalized to mean one; a Haar (tight) frame is the flat line. Middle and right: distribution of $\sqrt{n}\cos\theta$ for random pairs, raw, centred and whitened, against the Haar/Gaussian prediction (dashed). Whitening restores the bulk; the tails remain heavy.

they are the same word with and without a leading space or capital ("As"/" As", "Georgia"/" Georgia", "Education"/"education"), inflections ("turned"/"turning", "tend"/"tends") and near-synonyms ("spokesman"/"spokeswoman", "Usually"/"Normally"), with $|\cos| \approx 0.7$–0.9. In GloVe they are rare tokens of one type: numbers ("28.6"/"24.9", "1679"/"1677", "84-year"/"83-year"), Polish place names, moth genera. Either way they are not independent features at all. Nevertheless the bulk behaves like a random code: ℓ_1 recovery of random sparse combinations from a random sub-dictionary with $m = 2n$ sits at the Haar threshold.

Two cautions keep this last result in proportion. First, it is less surprising than it looks: the Donoho–Tanner transition is known to be universal well beyond Gaussian matrices, including for many deterministic frames [12], and we are seeing one more instance. Second, our test draws random supports, which is the typical case where agreement is expected. Real activations co-activate correlated features (the near-duplicates above), and a random-support test cannot see that. The embeddings tested are input embeddings; dictionaries of internal representations (sparse autoencoders on a residual stream) are the test that matters most and were not reachable here.

5.4. **What this means for the correspondence.** The Haar code is the right null model and the wrong description. Statements about *typical* sparse sets (recovery thresholds, the restricted isometry of random supports) transfer from Haar codes to trained ones, because they depend on second-order structure, which tight frames share. Statements about *worst-case* sets (coherence, the cheapest null vector) do not: trained codes can be better than random (toy models with plentiful features) or much worse (embeddings with near-duplicates). In black hole language, a trained code is a designed error-correcting code rather than a scrambler's typical one—which raises the question, not pursued here, of whether a black hole's actual code (as opposed to its Haar model) has analogous structure.

Part II. Legibility and accidental obfuscation

6. Setting: a channel, a public frame, and monitors

Part I was about the *geometry* of a code: its Gram matrix, which is invariant under rotations of the fundamental space. Monitoring depends on the part of the code that is *not* rotation invariant: how its features sit relative to a frame that an outside observer knows. Natural language is such a frame, and so is a sparse-autoencoder dictionary or the neuron basis. When

two models communicate in activations rather than tokens (latent reasoning [19], activation handoff between agents), the question is whether training keeps the message legible in the frame the overseer reads.

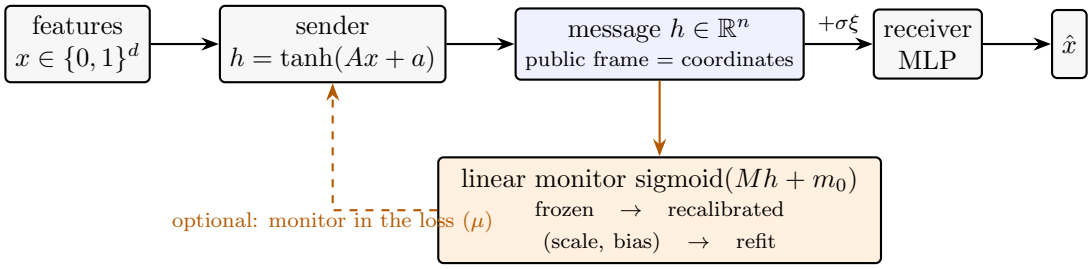


FIGURE 6. The toy model of Part II. A sender writes a message h in $\mathbb{R}^n$; the public frame is the coordinate basis, made meaningful by the coordinatewise nonlinearity. The receiver reads a noisy copy. A linear monitor reads h, either frozen at a legible checkpoint, recalibrated, or refit.

6.1. **The model.** Features $x \in \{0,1\}^d$ are i.i.d. Bernoulli(p). The sender writes $h = \tanh(Ax + a) \in \mathbb{R}^n$; the channel adds $\sigma\xi$ with $\xi \sim \mathcal{N}(0, I)$; a one-hidden-layer receiver outputs logits for x; the loss is the summed binary cross-entropy. Unless stated otherwise $n = 16$, $\sigma = 0.1$, $p = 0.1$, the receiver has 64 hidden units, and training uses minibatches of 256. A code is *legible* if each feature drives one public coordinate: $A_{jl} = 0$ unless $j = o(l)$, the coordinate it owns. We produce legible codes by training with an ℓ_1 budget $\lambda \sum_j |h_j|$ in the public frame, and then ask what happens when training continues.

6.2. **What we measure.** Two kinds of legibility, which turn out to behave differently.

- *Feature-level spread*: the participation ratio of each sender column, $\left(\sum_j A_{jl}^2\right)^2 / \sum_j A_{jl}^4$, averaged over features ("neurons per feature"; 1 for a legible code).
- *Monitor-level legibility*: the F1 score of linear monitors reading h. The *frozen* monitor is a logistic probe fit at the legible checkpoint and never touched again. The *recalibrated* monitor keeps the frozen directions and refits only a scale and bias per feature. The *refit* monitor is retrained on the current code with the same budget as the frozen one. The receiver's own F1 bounds what is in the channel.

The differences between consecutive levels in Fig. 7 separate three effects that earlier drafts of these notes conflated.

7. THEORY: THE LEGIBLE CRITICAL POINT

7.1. **What privileges the public frame.**

Proposition 7.1. *[proved]If the sender were linear, $h = Ax$, with an ℓ_2 power constraint, then for every orthogonal O the codes A and OA achieve the same optimal loss (the receiver absorbs $O^\top$, and the noise is isotropic). There is no preferred frame, and "legible" is not a meaningful property.*

The coordinatewise tanh breaks this symmetry, because it bounds each coordinate separately. This is the "privileged basis" of [2], and it has a second consequence. The feasible set of messages is a cube, and a message that uses k coordinates has energy at most k, whereas one that uses all n coordinates can have energy n while still respecting each bound. Against isotropic noise, energy buys distinguishability. So the same feature of the architecture that makes the public

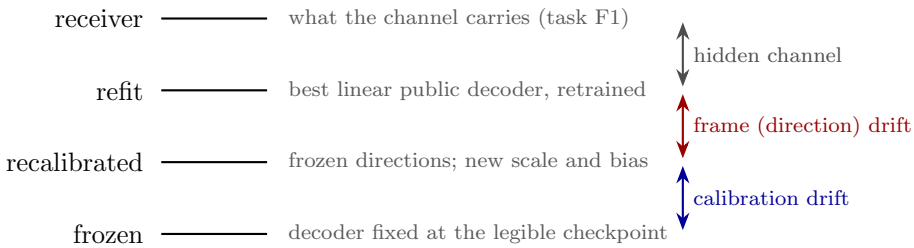


FIGURE 7. Four readings of the same message. Frozen → recalibrated measures calibration drift (amplitudes and offsets changed; cheap to repair with labels). Recalibrated → refit measures frame drift (the directions changed). Refit → receiver measures information the receiver uses that no linear public decoder recovers.

frame meaningful also makes spreading across it pay. (An earlier draft argued that spreading pays from energy considerations alone; without the per-coordinate bound that argument is false, since rotating a code changes nothing.)

7.2. **Stationarity in leak directions.**

Lemma 7.2. *[proved]Let the code be legible and let j be an* empty *coordinate (owned by no feature), so $h_j = \tanh(a_j)$ is constant. If the receiver ignores j ($B_{:,j} = 0$), then $\partial L/\partial A_{jl} = 0$ and $\partial^2 L/\partial A_{jl}^2 = 0$ for every feature l.*

Proof. $\partial L/\partial A_{jl} = \mathbb{E}\big[(\sum_u \delta_u B_{uj})\,\phi'(z_j)\,x_l\big]$ with $\delta = \partial L/\partial v$, which vanishes when $B_{:,j} = 0$. Differentiating again, every term carries a factor of $B_{:,j}$. □

At the receiver's optimum, reading a coordinate that carries only noise does not help, so one expects $B_{:,j} \approx 0$. (Trained receivers do not get there, because the cost of reading noise is second order and suppressed by the receiver's confidence. In practice the lemma holds in the weaker form that the leak gradient is proportional to $B_{:,j}$ times the residual error; see E13.) Leaks into a coordinate *owned by another feature* have no such protection. With more features than coordinates (superposition) every coordinate is owned, and a legible code need not be a critical point at all.

7.3. **Envelope curvature: the I-MMSE formula.** If the receiver re-optimizes as the sender leaks, the relevant quantity is the curvature of the envelope.

Proposition 7.3 (Envelope curvature)**.** *[proved] (for the Bayes receiver) Let ℓ own coordinate $o(\ell)$ and leak by ε into empty coordinate j, and let the receiver be Bayes-optimal. Then*

$$\left.\frac{d^2 L_{\mathrm{env}}}{d\varepsilon^2}\right|_{\varepsilon=0} = -\frac{\phi'(a_j)^2\,\mathrm{mmse}_\ell}{\sigma^2},$$

where $\mathrm{mmse}_\ell = \mathbb{E}(x_\ell - \mathbb{E}[x_\ell \mid y])^2$ is the residual error of feature ℓ and $\phi' = 1 - \tanh^2$.

Proof. Features are independent and each coordinate depends on one feature, so the loss for ℓ is $H(x_\ell) - I(x_\ell; y)$. The leak adds a second Gaussian observation of x_ℓ with gap $\phi'(a_j)\varepsilon$ and noise σ; the two observations combine into one with $\mathrm{snr} = (g_\ell^2 + \phi'(a_j)^2\varepsilon^2)/\sigma^2$. By the I-MMSE relation [17], $dI/d\,\mathrm{snr} = \frac{1}{2}\mathrm{mmse}$ for any input distribution, so $d^2 I/d\varepsilon^2|_0 = \phi'(a_j)^2 \mathrm{mmse}_\ell/\sigma^2$. □

The formula unifies two sources of negative curvature that earlier drafts treated separately. When a feature is read poorly because of interference, mmse_ℓ stays large and the force is $\propto 1/\sigma^2$.

When it is read poorly because of noise, mmse_ℓ is a Gaussian tail and the force is non-monotone in σ.

7.4. **What gradient descent actually sees: the handshake.** Gradient descent follows the joint Hessian, not the envelope. Restricted to the leak $\varepsilon = A_{j\ell}$ and the read $r = B_{:,j}$, Lemma 7.2 gives the block

$$\mathcal{H} = \begin{pmatrix} 0 & C^\top \\ C & H_{rr} \end{pmatrix}, \qquad \lambda_- = \tfrac{1}{2}\Big(h - \sqrt{h^2 + 4\|C\|^2}\Big) \approx -\|C\| \quad (h \ll \|C\|),$$

with h the receiver curvature along C [heuristic]. The sender cannot profit from leaking until the receiver reads, and the receiver cannot profit from reading until the sender leaks: the unstable direction is half leak, half read, and the envelope curvature is the Schur complement $-C^\top H_{rr}^{-1} C$. Under gradient descent with step η, a perturbation of size z_0 escapes in a time of order $\log(1/z_0)/(\eta|\lambda_-|)$, so a code whose features are already well read lingers at the saddle. Note what this implies at the extreme: every curvature in this picture carries a factor of the receiver's residual error (Proposition 7.3), so when the features are decoded almost perfectly the saddle is almost flat, and gradient descent does not leave it at all on any practical time scale.

Adam divides each coordinate's update by a running estimate of its gradient magnitude. Near the saddle that magnitude is tiny, and the normalized step has size $\approx \eta$ regardless of $\|C\|$. Two consequences follow:

(1) Adam's escape time is essentially independent of the curvature (Fig. 8).
(2) Even where the expected gradient vanishes, minibatch noise in the leak gradient is normalized into steps of size $\sim \eta$. So the sender random-walks off the legible support whether or not the receiver reads the result. The message densifies without being used.

The second mechanism needs no negative curvature at all. E13 below finds that it is the one operating at the noise level of our drift experiments, where the legible point is flat to numerical precision.

Prediction 7.4. Spread of the message in the public frame is controlled by the update rule of the parameters that *write* the message. Plain SGD on the sender preserves legibility even if everything else uses Adam; Adam on the sender spreads the message even if the receiver uses SGD.

8. Experiments: drift, optimizers, and the saddle

8.1. **E10: what happens to a legible code.** Starting from a legible code with $d = 32$ features in $n = 16$ coordinates, we continue training under Adam in four ways (Table 6, Fig. 9).

Release (the budget is removed and nothing else changes) is the cleanest case. The code spreads: neurons per feature rise from 1.16 to 1.73. The frozen monitor falls from 0.92 to 0.71. But a per-feature recalibration of scale and bias recovers most of that (0.87), and a refit monitor reads the new code as well as the old (0.94). The honest decomposition of the frozen monitor's loss is roughly two-thirds calibration and one-third direction. Neither is invisible: a monitor that is refit, or even just recalibrated, on fresh labelled data sees through it.

Pressure (three times as many active features) does not spread the code at the feature level; neurons per feature fall slightly. *Pressure with the budget kept* holds the code legible at a visible price in task performance (receiver F1 0.83 against 0.92): the legibility tax. *Pressure with the frozen monitor in the loss* does what one would expect of a training signal: the frozen monitor's score stops falling. The worry from earlier drafts was that the receiver would meanwhile build a channel no monitor can read. At this scale we see no sign of it: the receiver−refit gap is the same with and without the monitor in the loss, within seed-to-seed variation.

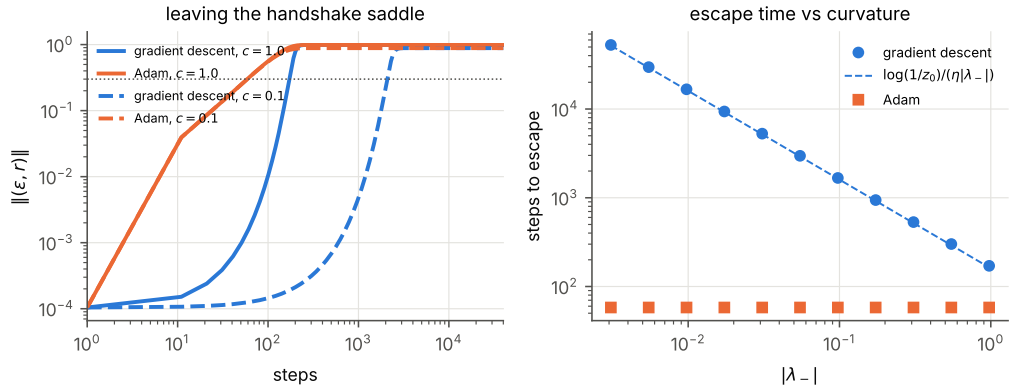


FIGURE 8. E14: the handshake saddle in isolation, $L = c\,\varepsilon r + \frac{h}{2}r^2 + \frac{q}{4}(\varepsilon^2 + r^2)^2$ with $h = 0.05c$, $q = c$. Left: distance from the saddle against steps; gradient descent waits, then grows exponentially; Adam (both values of c coincide) leaves at once. Right: escape time against $|\lambda_-|$ over two and a half decades. Gradient descent takes 1.6–4% longer than $\log(1/z_0)/(\eta|\lambda_-|)$ (the excess grows with $|\lambda_-|$, where the discrete-step and quartic corrections matter); Adam takes 58 steps throughout.

condition (after 6000 steps)	receiver F1	frozen	recalibrated	refit	neurons/feature
legible checkpoint	0.915 ± 0.018	0.922 ± 0.019	0.927 ± 0.015	0.922 ± 0.017	1.159 ± 0.123
release	0.984 ± 0.001	0.714 ± 0.047	0.865 ± 0.061	0.937 ± 0.013	1.733 ± 0.336
pressure	0.922 ± 0.006	0.790 ± 0.066	0.839 ± 0.053	0.893 ± 0.015	1.085 ± 0.055
pressure+budget	0.827 ± 0.012	0.772 ± 0.052	0.798 ± 0.026	0.822 ± 0.021	1.214 ± 0.216
pressure+monitor	0.907 ± 0.004	0.803 ± 0.018	0.872 ± 0.030	0.870 ± 0.011	1.241 ± 0.156

TABLE 6. E10: four continuations of a legible code ($d = 32$, $n = 16$, Adam), mean $\pm$ sd over 5 seeds. "pressure": feature density p raised from 0.1 to 0.3; "budget": the ℓ_1 legibility budget is kept; "monitor": the frozen monitor's loss is added to the training loss. Receiver$-$refit gap: 0.029 ± 0.019 under pressure, 0.037 ± 0.011 with the monitor in the loss.

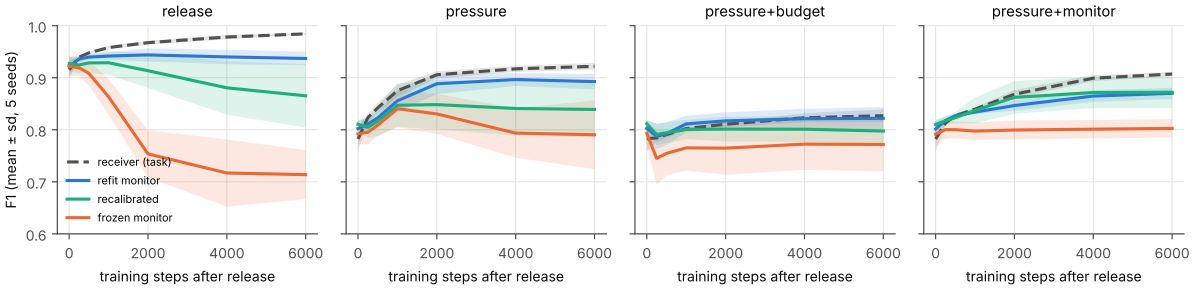


FIGURE 9. E10: the four readings of Fig. 7 over training, mean $\pm$ sd over 5 seeds.

8.2. **E11: the update rule.** We now hold the task fixed, remove the budget, and change only the update rule (Table 7, Fig. 10).

optimizer (6000 steps, budget removed)	receiver F1	frozen	recalibrated	refit	neurons/feature	leak
$d = 12$, legible start	0.985 ± 0.003	0.998 ± 0.004	0.998 ± 0.004	0.998 ± 0.004	1.033 ± 0.066	0.013 ± 0.026
Adam	1.000 ± 0.000	0.998 ± 0.004	0.998 ± 0.004	1.000 ± 0.000	2.066 ± 0.097	0.330 ± 0.016
SGD	1.000 ± 0.000	0.998 ± 0.003	0.998 ± 0.003	1.000 ± 0.000	1.107 ± 0.132	0.045 ± 0.052
hybrid: SGD sender, Adam receiver	1.000 ± 0.000	0.998 ± 0.003	0.998 ± 0.003	1.000 ± 0.000	1.078 ± 0.094	0.032 ± 0.038
reverse: Adam sender, SGD receiver	1.000 ± 0.000	0.998 ± 0.004	0.998 ± 0.004	1.000 ± 0.000	1.726 ± 0.215	0.315 ± 0.172
SGD, large step	0.856 ± 0.201	0.430 ± 0.142	0.516 ± 0.202	0.843 ± 0.211	3.884 ± 0.545	0.620 ± 0.164
$d = 32$, legible start	0.916 ± 0.017	0.927 ± 0.022	0.934 ± 0.020	0.926 ± 0.021	1.269 ± 0.172	0.071 ± 0.028
Adam	0.982 ± 0.002	0.708 ± 0.052	0.867 ± 0.070	0.941 ± 0.015	1.775 ± 0.432	0.069 ± 0.030
SGD	0.960 ± 0.004	0.796 ± 0.083	0.919 ± 0.039	0.943 ± 0.014	1.561 ± 0.350	0.102 ± 0.048
hybrid: SGD sender, Adam receiver	0.982 ± 0.002	0.760 ± 0.050	0.886 ± 0.055	0.944 ± 0.015	1.765 ± 0.401	0.098 ± 0.037
reverse: Adam sender, SGD receiver	0.962 ± 0.003	0.769 ± 0.081	0.916 ± 0.040	0.944 ± 0.013	1.547 ± 0.367	0.065 ± 0.033
SGD, large step	0.426 ± 0.357	0.232 ± 0.045	0.143 ± 0.195	0.426 ± 0.301	2.361 ± 0.842	0.687 ± 0.208

TABLE 7. E11: same task, legibility budget removed, five update rules; mean ± sd over 5 seeds. Learning rates: Adam 3×10^{-3}, SGD 0.05, large-step SGD 0.5. "leak": fraction of sender weight energy outside each feature's legible coordinate. $d = 12 < n$: empty coordinates exist; $d = 32 > n$: superposition.

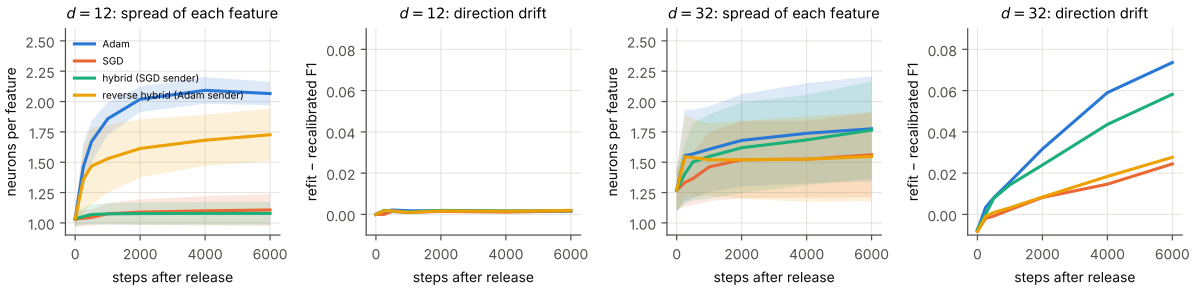


FIGURE 10. E11: feature spread and direction drift (refit minus recalibrated F1) after the budget is removed, for four update rules, mean ± sd over 5 seeds. Large-step SGD diverges on some seeds and is shown only in Table 7.

With spare public coordinates ($d = 12 < n$), Prediction 7.4 holds. At identical task performance (F1 1.000 for all four rules), the sender's update rule decides whether features stay on their coordinates. Adam on the sender spreads each feature over about two neurons, whether the receiver uses Adam (2.07) or SGD (1.73). SGD on the sender keeps it on one (1.11 with SGD everywhere, 1.08 for the hybrid with an Adam receiver). This is the hybrid optimizer of earlier drafts, and in this regime it works as advertised. The spread is also exactly the kind predicted by Section 7: under Adam on the sender, the message densifies even when the receiver (on SGD) is too slow to make use of it. In this regime the spread does not hurt linear monitors, which remain at 0.998 in every column: the new copies are redundant, and the original coordinate still carries each feature.

In superposition ($d = 32 > n$), Prediction 7.4 fails. E11's starting codes differ in how they are packed (`e11_packing.py`): at $d = 12$, 11 or 12 of the 16 coordinates are used and at most one is shared; at $d = 32$, 10 to 16 coordinates are shared by 2–4 features each, and at most two are empty. So Lemma 7.2 has almost nothing to protect, and E15 below shows that leaks into occupied coordinates feel first-order forces. Spread and direction drift now track the *receiver's* update rule: with an Adam receiver (Adam everywhere, or the hybrid) neurons per feature reach 1.77 and direction drift is 0.06–0.07; with an SGD receiver (SGD everywhere, or the reverse hybrid) they are 1.55 and 0.02–0.03. These are also the configurations whose receivers learn more (task F1 0.982 against 0.960). The natural reading is that, in superposition, drift away from

the public frame is a first-order response to a receiver that is still improving, and the optimizer matters through how fast the receiver improves rather than through the sender's normalization. The hybrid optimizer does not preserve legibility here at matched task performance.

8.3. **E15: superposition removes the critical point.** Is the legible code a critical point when coordinates are shared? We take E11's starting codes (same seeds, same ReLU receiver), restrict each feature to its strongest coordinate while keeping whatever sharing the code had, polish all parameters within that support on a full batch, and then measure the gradient in the directions that leave it (Table 8). At $d = 12$ the polish converges, and the legible code is a

		coordinates (of 16)		gradient within support		gradient, leaks into occupied		leaks into
d	seeds	empty	shared	norm	per entry	norm	per entry	empty, norm
12	3	4.3	0.3	1.9e-09–2.7e-07	4.3e-11–6.1e-09	7.7e-10–6.6e-07	6.4e-11–5.7e-08	2.7e-10–2.7e-07
32	3	0.3	13.7	2.9e-02–4.1e-02	5.1e-04–7.3e-04	1.5e-01–1.8e-01	7.0e-03–8.1e-03	0.0e+00–3.9e-04

TABLE 8. E15: is the legible code a critical point? Starting from E11's legible codes (same seeds, ReLU receiver), each feature is restricted to its strongest coordinate (shared coordinates kept) and all parameters are polished within that support (full batch of 6144, $\sigma = 0.1$). Gradient norms, and root-mean-square per entry, in three groups of directions; ranges are over seeds.

critical point in every direction, leaks included: all gradient norms lie between 10^{-9} and 10^{-6}. At $d = 32$ it is not. After the same 11,000 full-batch steps the gradient within the legible support is still 3–4 $\times 10^{-2}$, and the gradient for leaks into occupied coordinates is 0.15–0.18 in norm, 11–14 times larger per entry. In superposition the legible support is not where the loss wants to be. Drift away from it is a first-order response, which any optimizer will follow, and which makes the update rule matter mainly through how fast the receiver keeps improving. That is the pattern E11 shows.

8.4. **E13: how flat is the legible critical point?** To look for the saddle of Section 7 directly we use $d = 12$ features (so empty coordinates exist), a smooth softplus receiver (so second derivatives exist), and channel noise from the operating value $\sigma = 0.1$ up to $\sigma = 0.9$. We match each feature to its own coordinate, restrict the sender to that legible support, and polish everything to a critical point within it, on a fixed batch of 6144 samples. We then compute the global smallest Hessian eigenvalue by Lanczos and, for the least saturated empty coordinate, the leak/read block of each feature (Table 9, Fig. 11). Three findings, one of them negative.

At the operating point the legible code is flat, not a saddle. At $\sigma = 0.1$, the noise level of E10 and E11, the polished point is critical to 6×10^{-9}, and the global $\lambda_{\min}$, the leak curvatures and the envelope curvatures are all below 10^{-9} in magnitude. This is not an artefact. The receiver's error on fresh noise is itself a few times 10^{-11}, so Proposition 7.3 predicts a negative curvature of order 10^{-12}. Every second derivative of the cross-entropy carries a factor of the receiver's residual error, and a receiver that decodes almost perfectly sits on a plateau. There is no saddle to escape. Gradient descent has nothing to follow, while Adam turns vanishing gradients into steps of size η. This is the second mechanism of Section 7, and it is the one that accounts for E11 at $d = 12$: Adam on the sender spreads the code, the spread buys nothing on the task, and SGD on the sender leaves the code where it is.

The receiver does not ignore empty coordinates. Lemma 7.2 assumes that the receiver puts no weight on a coordinate that carries only noise. Trained receivers do: the weight on the empty coordinate we analyse is comparable to the weight on owned coordinates at every σ (`e13_fresh.py`). Its cost is second order in σ and suppressed by the receiver's confidence, so it

	sender	residual error mmse		gradient norm		global $\lambda_{\min}$		leak into empty coordinate (median over features)		
σ	gap	batch	fresh noise	within	leak, empty	value	on leak	$H_{\varepsilon\varepsilon}$ (lemma: 0)	envelope	I-MMSE
0.1	1.27	4e-18	3e-11	6e-09	4e-10	-5e-10	0.04	1e-10	1e-10	-6e-13
0.2	1.61	2e-17	4e-05	7e-09	1e-09	-1e-09	0.69	8e-10	4e-10	-8e-05
0.4	1.97	7e-14	8e-03	1e-07	9e-08	-6e-08	0.38	3e-07	2e-07	-2e-03
0.6	2.00	1e-02	3e-02	1e-03	2e-02	-8e-04	0.07	1e-01	1e-01	-5e-02
0.9	2.00	4e-02	6e-02	2e-03	2e-02	-2e-02	0.85	5e-02	1e-02	-6e-02

TABLE 9. E13: the legible critical point ($d = 12$, $n = 16$, softplus receiver), polished on a fixed batch of 6144 samples with fixed channel noise. Sender gap: $|\tanh(A_{o\ell} + a_o) - \tanh(a_o)|$, at most 2. Residual error of the trained receiver on that batch and on fresh noise (`e13_fresh.py`). Gradient norm within the legible subspace and for leaks into empty coordinates. Global $\lambda_{\min}$ by Lanczos and the weight of its eigenvector on leaks into empty coordinates. Last three columns: for the least saturated empty coordinate, the leak curvature $H_{\varepsilon\varepsilon}$, the envelope (Schur-complement) curvature, and the I-MMSE prediction $-\phi'(a_j)^2 \mathrm{mmse}_\ell/\sigma^2$ of Proposition 7.3 with the fresh-noise error.

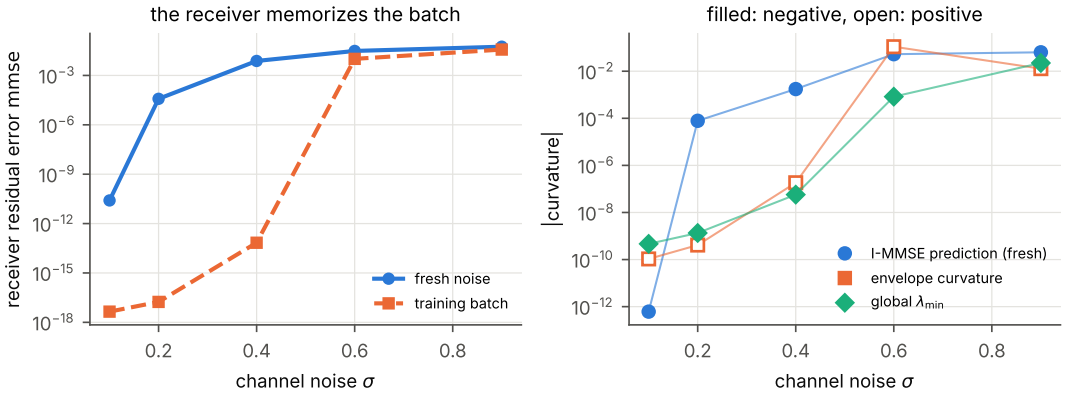


FIGURE 11. E13. Left: residual error of the polished receiver on its training batch and on fresh channel noise; for $\sigma \leq 0.4$ the receiver has memorized the noise in its batch. Right: magnitudes of the I-MMSE prediction (with the fresh-noise error), of the measured envelope curvature (median over features), and of the global smallest Hessian eigenvalue; filled markers are negative, open markers positive.

survives from initialization. The lemma therefore holds only in a weaker form: the leak gradient is proportional to that weight times the residual error, which is negligible at the operating point (E15, $d = 12$) but not at high noise.

The quantitative curvature theory is not confirmed. At $\sigma = 0.2$ and 0.4 everything is again flat, but for a different reason: on its fixed batch the receiver has memorized the channel noise (residual error 10^{-17} and 10^{-13} on the batch, against 4×10^{-5} and 8×10^{-3} on fresh noise). The Hessian we compute is the Hessian of an overfit empirical loss, and the I-MMSE prediction with the fresh-noise error (-8×10^{-5}, -2×10^{-3}) is orders of magnitude larger than anything measured. At $\sigma = 0.6$ and 0.9 the batch and fresh errors are closer, and negative curvature appears: the global $\lambda_{\min}$ is -8×10^{-4} and -2×10^{-2}, and at $\sigma = 0.9$ its eigenvector lies 85% on leaks into empty coordinates. So the legible code is a saddle at high noise. But it is not the

saddle of Section 7. The point is not critical in leak directions (gradient about 2×10^{-2}, against 1.5–2.5 $\times 10^{-3}$ within the support), the leak curvature is positive rather than zero because the receiver reads the empty coordinate, and the per-feature envelope curvature is positive (for 23 of the 24 features) where Proposition 7.3 predicts it negative. A clean test of the I-MMSE formula for trained receivers needs the Hessian of the population loss (fresh noise and much larger batches), which we have not done.

9. What this means for monitoring

We summarize what the toy model supports, what it does not, and what would test it at scale.

Supported.

(1) *Frozen monitors decay; most of the decay is cheap to repair.* Removing a legibility pressure lets the code move, and a frozen linear monitor loses about 0.2 F1. Recalibrating a scale and bias per feature, with fresh labels, recovers most of it. A monitor of a channel under training should be recalibrated on a schedule, not frozen.

(2) *Where the public frame has spare capacity, the update rule of the writer decides legibility.* Plain SGD on the parameters that write the message keeps features on their coordinates at no cost to the task, even with Adam elsewhere. The mechanism is that near a well-trained legible code the loss is flat (E13), and Adam normalizes the resulting tiny and noisy gradients into finite steps.

(3) *Legibility can be bought, at a price.* An ℓ_1 budget in the public frame holds the code legible under pressure and costs task performance.

Not supported, or not tested.

(1) *Spontaneous obfuscation under monitor pressure.* Putting a frozen monitor in the loss did not open a gap between what the receiver uses and what a refit linear monitor reads. Obfuscation under monitor pressure has been observed in reinforcement learning on chain-of-thought with large models [20]; this toy is too small and too linear to say anything about that.

(2) *Optimizer control in superposition.* When the frame has no spare capacity, drift is first-order (E15), and we found no update rule that both keeps the receiver improving and keeps the code legible.

(3) *The curvature theory.* The I-MMSE formula for the curvature at the legible code is proved for a Bayes receiver and untested for trained ones (E13).

(4) *Anything at scale.* All of Part II is a 16-dimensional channel.

The reassuring disanalogy. No-cloning has no classical counterpart (§4). Whatever obstructs a monitor from reading a classical channel is ignorance of the code or the cost of decoding, never a law of nature. Every effect in Part II was visible to a monitor that was allowed to refit.

Predictions for latent-reasoning models. The natural test bed is continuous chain-of-thought [19]: a model feeds its own hidden state back as the next input, so it is simultaneously the sender and the receiver of an activation-space message. Starting from a text-CoT checkpoint (legible by construction in the vocabulary frame, or in a sparse-autoencoder frame), the notes make these predictions.

Prediction 9.1. Logit-lens and SAE-frame legibility of the latent thoughts decays when the text bottleneck is removed, and the decay is mostly recoverable by recalibrating a frozen probe (E10 at scale).

Prediction 9.2. If the frame has spare capacity (features the message is not yet using), training the parameters that write the latent thought with plain SGD, and everything else with AdamW,

slows the spread of the message relative to AdamW everywhere, at matched task accuracy. If the frame is saturated, it does not (E11 at scale).

Prediction 9.3. Near a well-trained legible checkpoint with spare frame capacity, the expected gradient of the writer's parameters in directions that leave the frame is close to zero, while AdamW's update along those directions is as large as along any other (E13 and E15 at scale).

These need a model of about 10^8 parameters and tens of GPU-hours; none has been run.

10. Status ledger

claim	status	where
Random codes over $\mathbb{R}$ and $\mathbb{C}$ are isometries on sets with $\log\|\Sigma\| \ll \beta N\delta^2$	proved	Theorem 3.6
AEHPV low-complexity isometry and RIP are two cases of it	proved	Corollaries 3.7 and 3.8
Exact overlap variances; real/complex difference is the Ω Brauer term	proved, checked	Proposition 3.2, E01
Any unit-norm tight frame matches Haar at second order	proved	Proposition 3.3
State-independent reconstruction is biased by $\approx e^{-S_{\mathrm{BH}}}\,\mathrm{tr}\,A$	proved	Proposition 4.3
Page $\leftrightarrow$ Donoho–Tanner; QES crossover $\leftrightarrow$ ALMT edge	proved (each side), structural analogy	§4.3
Python's lunch $\leftrightarrow$ computational–statistical gap	analogy (hardness conjectural)	§4.3
Trained toy codes are tight, Haar at 2nd order, better than Haar at the extremes	checked	E02
Language-model embeddings: anisotropic, heavy-tailed, Haar-like recovery	checked	E03
Legible codes are critical in leaks to empty coordinates (if the receiver ignores them)	proved	Lemma 7.2
… and, from E11's starting codes, critical at $d = 12$ but not at $d = 32$	checked	E15
Envelope curvature $= -\phi'^2\mathrm{mmse}/\sigma^2$ for the Bayes receiver	proved	Proposition 7.3
… and for trained receivers	not confirmed	E13
At the operating noise the legible code is flat ($\|\lambda\| < 10^{-9}$)	checked	E13
Gradient descent escapes in $\sim \log(1/z_0)/\eta\|\lambda_-\|$, Adam in $O(1)$	heuristic; checked in isolation	E14
Writer's update rule controls spread when the frame has spare capacity	checked	E11, $d = 12$
… and not in superposition	checked (negative)	E11, $d = 32$
Monitor in the loss grows a hidden channel	not supported at this scale	E10
Anything about large models	untested; predictions only	§9

Appendix A. Reproducing these notes

Everything is pure Python (numpy, scipy, matplotlib, msgpack); no GPU, no deep-learning framework. On a 2-core machine the full pipeline takes about two hours, dominated by E11, E13 and the RoBERTa linear programs in E03.

step	command	output
gradient check	`python -m tests.test_gradients`	stdout
fetch weights	`bash scripts/fetch_weights.sh`	`data/`
E01 exact moments	`python -m experiments.e01_haar_moments`	`results/e01_*.json`
E02 toy models	`python -m experiments.e02_tms_vs_haar`	`results/e02_*`
E03 embeddings	`python -m experiments.e03_real_embeddings`	`results/e03_*.json`
E03 coherent pairs	`python -I scripts/e03_top_pairs.py`	`results/e03_top_pairs.json`
E10 drift conditions	`python -m experiments.e10_drift_conditions`	`results/e10_*.json`
E11 optimizers	`python -m experiments.e11_optimizers`	`results/e11_*.json`
E11 packing	`python -m experiments.e11_packing`	`results/e11_packing.json`
E13 saddle Hessian	`python -m experiments.e13_saddle_hessian`	`results/e13_*.json`
E13 fresh-noise check	`python -m experiments.e13_fresh`	`results/e13_fresh.json`
E14 handshake toy	`python -m experiments.e14_bilinear_handshake`	`results/e14_*.json`
E15 forces in superposition	`python -m experiments.e15_superposition_forces`	`results/e15_*.json`
figures, tables	`python scripts/make_figures.py; python scripts/make_tables.py`	`notes/figs, notes/tables`

Data. RoBERTa-base token embeddings come from spaCy's `en_core_web_trf-3.8.0` wheel (GitHub release `explosion/spacy-models`). The weights live in a PyTorch pickle inside a msgpack blob. We read it without PyTorch, using an unpickler that resolves only a whitelist of tensor-rebuild functions, so no code from the file is executed (`scripts/extract_roberta_embeddings.py`). Nearest neighbours confirm the extraction ("king" → "queen", "Paris" → "France", "quantum" → "photon"). GloVe 100d is the `glove-wiki-gigaword-100` release of `RaRe-Technologies/gensim-data`. GPT-2 and sparse-autoencoder dictionaries, the natural next tests, were not reachable from the environment these notes were produced in.

Conventions. All F1 scores are micro-averaged over features. Every stochastic experiment fixes its seeds in the script; results reported as mean ± sd are over the seeds listed in the corresponding table.

Appendix B. Corrections to earlier drafts

These notes grew out of conversations in which several claims were made before they were tested. We list the ones that changed, because the changes are part of the result.

earlier claim	what we now think	evidence
Removing the legibility budget collapses a frozen monitor ($0.92 \to 0.64$, one seed).	The drop is real ($0.92 \to 0.71 \pm 0.05$, five seeds) but mostly *calibration*: refitting a scale and bias per feature restores 0.87; full refit gives 0.94.	E10
Task pressure densifies the code.	Not at the feature level: neurons per feature go from 1.16 to 1.09 when p triples. Message-level density rises only because more features are active.	E10
A monitor in the loss keeps its score while a hidden channel grows.	No evidence at this scale: the receiver−refit gap is 0.037 ± 0.011, against 0.029 ± 0.019 without the monitor.	E10
Dense codes win because they carry more energy (n vs k).	Only with per-coordinate bounds (here tanh); for a linear sender, rotation symmetry makes the argument void.	§7
Frame rotations are exactly flat directions.	Exact only for a linear sender; the coordinatewise nonlinearity breaks the symmetry, which is also what makes the public frame meaningful.	§7
Global $\lambda_{\min} \approx -2 \times 10^{-4}$ at the legible point.	That number came from an unconverged shifted power iteration, and finite-difference Hessians through ReLU kinks are unreliable anyway. Replaced by Lanczos at a constrained critical point with a smooth receiver.	E13
The legible code is a saddle.	Only critical where coordinates are unshared; in superposition, leaks into occupied coordinates feel first-order forces.	E15
Adam leaves the legible code because it is a saddle, escaping at a rate independent of the curvature.	At the operating noise the curvature is below 10^{-9}: the code sits on a plateau, and Adam moves because it normalizes vanishing gradients, not because it escapes along a negative direction.	E13
The curvature at the legible code is the I-MMSE value ("checked" in an earlier version of these notes).	Not confirmed. The first measurement analysed a saturated coordinate; the corrected one is confounded by the receiver memorizing its fixed noise sample and by its weight on empty coordinates.	E13
Unifying theorem stated via Gordon's theorem over $\mathbb{F}$.	Complex Gordon is not needed; the finite-set theorem is proved for both fields via the Beta law of a compressed norm.	Theorem 3.6

earlier claim	what we now think	evidence
Adam versus SGD decides whether a code stays legible (one seed, $d = 12$).	Confirmed over five seeds when the frame has empty coordinates, and refined: it is the *sender's* rule that matters. False in superposition.	E11
A hybrid optimizer (SGD on the channel parameters, Adam elsewhere) keeps legibility at matched task performance.	True for $d = 12 < n$ (neurons per feature 1.08 vs 2.07 for Adam, task F1 1.000 for both). Not true for $d = 32 > n$, where the hybrid spreads as much as Adam.	E11

Appendix C. How these notes were made, in more detail

September 2026. D.W. proposed the correspondence between the AEHPV non-isometric code and superposition, and asked for the mathematics and for its bearing on governing agents that communicate in activations. In conversation, Claude proposed the unifying width condition, the reconstruction dictionary, a first numpy toy model of legibility drift, and the saddle and

I-MMSE analysis; D.W. chose to pursue the monitorability direction and asked for the analytics and checks. On 24 September Claude wrote a first set of LaTeX notes on Part I at D.W.'s request.

September–October. D.W. decided to split the work into a formal paper and a monitorability paper, chose titles, and drafted introductory text. The first paragraph of Section 1 is D.W.'s, lightly edited at his request; the black hole background in Section 2.1 is a rewrite by Claude of D.W.'s draft. D.W. put the project on hold because he did not want to believe the correspondence before testing it on trained networks. That is the origin of Section 5.

November 2026. In a single working session, Claude wrote the repository, ran every experiment reported here, and drafted these notes, with D.W. setting the scope ("working notes, transparent about how they were made"). During that session the experiments overturned several earlier claims (Appendix B). The model was Claude Opus 5.5.

What has and has not been checked. Every number is produced by code in the repository, and the hand-written gradients pass a finite-difference test. The proofs marked [proved] were written by Claude; at the time of this version D.W. has not checked all of them, and this paragraph will say so explicitly when he has.

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